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32. Least Recently Used (LRU) Page Replacement

Code:

```
#include <stdio.h>

int main() {    int pages[20],
frames[10], time[10];    int n, f, faults
= 0, counter = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(int i = 0; i < n; i++)        scanf("%d",
&pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(int i = 0; i < f; i++)
frames[i] = -1;

    for(int i = 0; i < n; i++)
    {        int found
= -1;

        for(int j = 0; j < f; j++)
if(frames[j] == pages[i])
found = j;

        if(found != -1)
time[found] = ++counter;
else        {        int pos = 0;

            for(int j = 1; j < f; j++)
if(frames[j] == -1 ||
time[j] < time[pos])        pos =
j;
```

```
        frames[pos] = pages[i];
time[pos] = ++counter;
faults++;
    }
}

printf("Total Page Faults = %d\n", faults);

return 0;
```

} **Output:**

```
Enter number of pages: 8
Enter page reference string:
1 2 3 4 4 3 2 1
Enter number of frames: 3
Total Page Faults = 7

...Program finished with exit code 0
Press ENTER to exit console.
```